CanadianSolar three-phase Inverter Modbus/RTU Communication Protocol

V20231103

Contents

1. Overview 1
2. Modbus/RTU Communication protocols 1
3. Communication protocols content 1
    3.1 Protocol Frame Format 1
    3.2 Address code 1
    3.3 Function Code 2
    3.4 Data Area 2
    3.5 Verification Code 2
4. Protocol Content 2
    4.1 Input register table - Grid status information data area 3
    4.2 Input Register table - Inverter (AC) output status information data area 6
    4.3 Input Register Table - Inverter (PV) input status information data area 7
    4.4 Input register table - Inverter internal status information data area 9
    4.5 Input register table - Inverter fault status information data area 11
    4.6 Holding register table -AC Protection Parameter Area 11
    4.7 Holding register table -Active Power Derating Parameter Area 13
    4.8 Holding register table -Reactive Power Derating Parameter Area 15
    4.9 Holding register table -Other Parameter Area 16
    4.10 Holding register table -Enable Control Parameter Area 18
    4.11 Holding register table -Control Command Area 19
    4.12 Holding register table -Inverter Basic Information Area DSP 20
    4.13 Holding register table -Basic Information area for ARM 21
5. Fault Analysis 22
6. CRC16-RTU verification code 27
7. Modbus address assignment 28
    7.1 Address Search Command(Broadcast) 28
    7.2 Address Assignment Command 29
    7.3 Unregistering Address Command(Broadcast) 30
8. Example 30
    8.1 Read one input register instruction. 30
    8.2 Read multiple input registers instruction 30
    8.3 Read one hold register instruction 31
    8.4 Read multiple hold registers instruction 31
    8.5 Write a hold register instruction 31
    8.6 Write multiple hold registers instruction 31
    8.7 Modbus address assignment 32
        8.7.1 Flow chart 32
        8.7.2 Example 32

1. Overview

This document describes the Modbus/RTU communication protocol of CanadianSolar three-phase grid-connected photovoltaic inverter only.

2. Modbus/RTU Communication protocols

- Communication interface, RS485;
- Communication parameters: 9600Bauds, 8N1;
- Check way: CRC16 - RTU;
- Data transmission formats: frame, 5 ~ 256 bytes;
- Data format: Big endian;

3. Communication protocols content

3.1 Protocol Frame Format

Table 1 Protocol frame format

Start	Address	Function	Data	CRC16-RTU	End
3.5 Char time	1 Byte	1 Byte	N Bytes	2 Byte	3.5Char time

3.2 Address code

The address code is the first byte of communication transmission. Valid slave device address ranges from 0 to 247, and the address range of each slave device ranges from 1 to 247. The host puts the slave address into the address code area of the information frame and addresses it to the slave. When the slave responds, it puts its own address into the address code area of the response information, so that the host can identify the slave address that responds. Address 0 is a broadcast address, which can be identified by all slaves but does not require a reply to the host.

3.3 Function Code

When the host sends a message to the slave, the function code tells the slave the action to be performed. When the slave responds to the host, the function code can indicate that the slave responds normally or an error occurs. In normal response, the slave replies data according to the corresponding function code; In case of an abnormal response, the slave machine makes an abnormal reply (reply function code +0x80).
Legal function codes are defined in the following table:

Table 2 Definition of function codes

Function Code	Function Description	Operation register
0x03	read holding registers	holding registers
0x04	read input registers	input registers
0x06	write holding register	holding registers
0x10	write multiple holding registers	holding registers

3.4 Data Area

The information data sent by the host to the slave device contains the request actions specified in the function code of the slave machine. If no error occurs, the response message from the slave machine to the host contains the request data. If there is an error, the slave machine does not respond.

3.5 Verification Code

CRC16-RTU verification mode is adopted using the standard MODBUS-RTU protocol.

4. Protocol Content

In Modbus protocol, the address range of the hold register is 0x0000~0x7FFF; The high 8 bits of its address represent the block address and the low 8 bits represent the register address within the corresponding block. As shown in the following table, the hold register is allocated into several blocks according to the data category, and the address range of each block is specified in the table.

Table 3 Input register block address allocation table

Address range	Data category
0x8000~0x80FF	Grid status Information data area
0x8100~0x81FF	Inverter (AC) Output status information Data area
0x8200~0x82FF	Inverter (PV) Input status information data area
0x8300~0x83FF	Data area of Inverter Internal Status Information
0x8400~0x84FF	Inverter Fault Status Information Data area

Hold register block address allocation table

0x0000~0x00FF	Mains Fault Protection Parameter Area
0x0100~0x01FF	Active Power Derating Parameter Area
0x0200~0x02FF	Reactive Power Derating Parameter Area
0x0500~0x05FF	Other Parameter Area
0x0600~0x06FF	Enable Control Parameter Area
0x0700~0x07FF	Control Command Area
0x0900~0x09FF	Inverter Basic Information Area-DSP
0x0B00~0x0BFF	Inverter Basic Information Area-ARM

4.1 Input register table - Grid status information data area

Table 4 Grid status information data area

Address	Variable description	Ratio	Unit	R/W	Data type	Remarks
0x8000	Line voltage Uab	0.1	V	R	S16	
0x8001	Line voltage Ubc	0.1	V	R	S16	

0x8002	Line voltage Uca	0.1	V	R	S16	
0x8003	A-phase voltage Ua	0.1	V	R	S16	
0x8004	B-phase voltage Ub	0.1	V	R	S16	
0x8005	C-phase voltage Uc	0.1	V	R	S16	
0x8006	A-phase grid frequency	0.1	Hz	R	S16	
0x8007	B-phase grid frequency	0.1	Hz	R	S16	
0x8008	C-phase grid frequency	0.1	Hz	R	S16	
0x8009	Grid phase sequence	1	NA	R	S16	0：NA 1：positive sequence 2：Reverse sequence
0x800A	Power grid voltage unbalance degree	0.1	%	R	S16	
0x800B	Grid system frequency	0.1	Hz	R	S16	50 Hz or 60Hz
0x800C	Voltage between N line and PE ground	1	V	R	S16	
0x800D ~ 0x8012	-					
0x8013	Daily power generation	0.1	kwh	R	U16	
0x8014	Daily generating time	0.1	h	R	U16	
0x8015	Total power generation (MSB)	1	kwh	R	U16	Merge display
0x8016	Total power output (LSB)				U16	
0x8017	Total generating time (MSB)	1	h	R	U16	Merge display
0x8018	Total generating time (LSB)				U16	
0x8019	The meter enables the flag bit	1		R	U16	
0x801A	Meter phase A voltage	0.1	V	R	S16	
0x801B	Meter phase B voltage	0.1	V	R	S16	
0x801C	Meter phase C voltage	0.1	V	R	S16	
0x801D	Meter phase A current	0.01	A	R	S16	

0x801E	Meter phase B current	0.01	A	R	S16	
0x801F	Meter phase C current	0.01	A	R	S16	
0x8020	Meter frequency	0.01	Hz	R	S16	
0x8021	A-B line voltage of the meter	0.1	V	R	S16	
0x8022	C-B line voltage of the meter	0.1	V	R	S16	
0x8023	A-C line voltage of the meter	0.1	V	R	S16	
0x8024	Meter phase A active power	0.1	kW	R	S16	
0x8025	Meter phase B active power	0.1	kW	R	S16	
0x8026	Meter phase C active power	0.1	kW	R	S16	
0x8027	Total active power of the meter	0.1	kW	R	S16	
0x8028	Meter phase A reactive power	0.1	kVAr	R	S16	
0x8029	Meter phase B reactive power	0.1	kVAr	R	S16	
0x802A	Meter phase C reactive power	0.1	kVAr	R	S16	
0x802B	Total reactive power of the meter	0.1	kVAr	R	S16	
0x802C	Meter phase A apparent power	0.1	kVA	R	S16	
0x802D	Meter phase B apparent power	0.1	kVA	R	S16	
0x802E	Meter phase C apparent power	0.1	kVA	R	S16	
0x802F	Total apparent power of the meter	0.1	kVA	R	S16	
0x8030	Meter phase A power factor	0.001		R	S16	
0x8031	Meter phase B power factor	0.001		R	S16	
0x8032	Meter phase C power factor	0.001		R	S16	
0x8033	Total power factor of the meter	0.001		R	S16	
0x8034	(current) Meter net Forward active energy (MSB)	0.1	kWh	R	U16	
0x8035	(current) Meter net Forward active energy (LSB)	0.1	kWh	R	U16	

0x8036	(current) Meter total Reverse active energy (MSB)	0.1	kWh	R	U16	
0x8037	(current) meter total Reverse active energy (LSB)	0.1	kWh	R	U16	
0x8038	Month power generation (MSB)	0.1	kWh	R	U16	Calendar month
0x8039	Month power generation (LSB)	0.1	kWh	R	U16	
0x803A	Year power generation (MSB)	0.1	kWh	R	U16	Calendar year
0x803B	Year power generation (LSB)	0.1	kWh	R	U16	
0x803C	Week power generation MSB)	0.1	kWh	R	U16	Calendar week
0x803D	Week power generation (LSB)	0.1	kWh	R	U16	

4.2 Input Register table - Inverter (AC) output status information data area

Table 5 Inverter (AC) output status information data area

Address	Variable description	Ratio	Unit	R/W	Data type	Remarks
0x8100	Phase A current Ia	0.1	A	R	S16	
0x8101	Phase B current Ib	0.1	A	R	S16	
0x8102	Phase C current Ic	0.1	A	R	S16	
0x8103	Phase A active power	0.1	KW	R	S16	
0x8104	Phase B active power	0.1	KW	R	S16	
0x8105	Phase C active power	0.1	KW	R	S16	
0x8106	Total active power of three phases	0.1	KW	R	S16	

0x8107	Phase A reactive power	0.1	kVAr	R	S16	
0x8108	Phase B reactive power	0.1	kVAr	R	S16	
0x8109	Phase C reactive power	0.1	kVAr	R	S16	
0x810A	Total reactive power of three phases	0.1	kVAr	R	S16	
0x810B	Phase A power factor	0.01	NA	R	S16	
0x810C	Phase B power factor	0.01	NA	R	S16	
0x810D	Phase C power factor	0.01	NA	R	S16	
0x810E	Three-phase power factor	0.01	NA	R	S16	
0x810F	Inverter A phase voltage	0.1	V	R	S16	
0x8110	Inverter B phase voltage	0.1	V	R	S16	
0x8111	Inverter C phase voltage	0.1	V	R	S16	

4.3 Input Register Table - Inverter (PV) input status information data area

Table 6 Inverter (PV) input status information data area

Address	Variable description	Ratio	Unit	R/W	Data type	Remarks
0x8200	PV connection mode	1	NA	R	S16	1- independent 2- Parallel
0x8201	PV Total input power	0.1	KW	R	S16	
0x8202	PV1 voltage	0.1	V	R	S16	
0x8203	PV1 current	0.1	A	R	S16	
0x8204	PV2 voltage	0.1	V	R	S16	

0x8205	PV2 current	0.1	A	R	S16	
0x8206	PV3 voltage	0.1	V	R	S16	
0x8207	PV3 current	0.1	A	R	S16	
0x8208	PV4 voltage	0.1	V	R	S16	
0x8209	PV4 current	0.1	A	R	S16	
0x820A	PV5 voltage	0.1	V	R	S16	
0x820B	PV5 current	0.1	A	R	S16	
0x820C	PV6 voltage	0.1	V	R	S16	
0x820D	PV6 current	0.1	A	R	S16	
0x820E	PV7 voltage	0.1	V	R	S16	
0x820F	PV7 current	0.1	A	R	S16	
0x8210	PV8 voltage	0.1	V	R	S16	
0x8211	PV8 current	0.1	A	R	S16	
0x8212	PV9 voltage	0.1	V	R	S16	
0x8213	PV9 current	0.1	A	R	S16	
0x8214	PV10 voltage	0.1	V	R	S16	
0x8215	PV10 current	0.1	A	R	S16	
0x8216	PV11 voltage	0.1	V	R	S16	
0x8217	PV11 current	0.1	A	R	S16	
0x8218	PV12 voltage	0.1	V	R	S16	
0x8219	PV12 current	0.1	A	R	S16	
0x821A	PV13 voltage	0.1	V	R	S16	
0x821B	PV13 current	0.1	A	R	S16	
0x821C	PV14 voltage	0.1	V	R	S16	
0x821D	PV14 current	0.1	A	R	S16	
0x821E	PV15 voltage	0.1	V	R	S16	

0x821F	PV15 current	0.1	A	R	S16	
0x8220	PV16 voltage	0.1	V	R	S16	
0x8221	PV16 current	0.1	A	R	S16	
0x8222	PV17 voltage	0.1	V	R	S16	
0x8223	PV17 current	0.1	A	R	S16	
0x8224	PV18 voltage	0.1	V	R	S16	
0x8225	PV18 current	0.1	A	R	S16	
0x8226 - 0x8231	-					
0x8232	Boost1 Current	0.1	A	R	S16	
0x8233	Boost2 Current	0.1	A	R	S16	
0x8234	Boost3 Current	0.1	A	R	S16	
0x8235	Boost4 Current	0.1	A	R	S16	
0x8236	Boost5 Current	0.1	A	R	S16	
0x8237	Boost6 Current,	0.1	A	R	S16	
0x8238	Boost7 Current	0.1	A	R	S16	
0x8239	Boost8 Current	0.1	A	R	S16	
0x823A	Boost9 Current	0.1	A	R	S16	

4.4 Input register table - Inverter internal status information data area

Table7 Data area of inverter internal status information

Address	Variable description	Ratio	Unit	R/W	Data type	Remarks
0x8300	Inverter operation status	1	NA	R	S16	0 - PowerOnMode 1 - PreCheckMode 2 - InvRunMode 3 - FaultMode 4 - ShutMode

						5 - BootloaderMode
0x8301	Inverter module temperature	0.1	℃	R	S16	
0x8302	BOOST module temperature	0.1	℃	R	S16	
0x8303	Internal temperature 1 (main ambient temperature)	0.1	℃	R	S16	
0x8304 - 0x8305	-					
0x8306	Insulation impedance detection value (ISO)	1	KΩ	R	S16	
0x8307	Leakage current detection value (GFCI)	1	mA	R	S16	
0x8308	Phase A DC current (DCI)	1	mA	R	S16	
0x8309	Phase B DC current (DCI)	1	mA	R	S16	
0x830A	Phase C DC current (DCI)	1	mA	R	S16	
0x830B	Positive BUS voltage	1	V	R	S16	
0x830C	Negative BUS voltage	1	V	R	S16	
0x830D	Positive and negative BUS voltage	1	V	R	S16	
0x830E	Power-on countdown	1	s	R	S16	
0x830F	ISO sampling circuit measured voltage	1	V	R	S16	
0x8310	Derating flag bit	1		R	S16	For debug
0x8311 - 0x831A	-					
0x831B	Export limit flag	1		R	S16	0:no limit 1:Export Reflux Limit 2:Export Reflux Disable Limit

4.5 Input register table - Inverter fault status information data area

Table 8 Inverter fault status information data area

Address	Variable description	Ratio	Unit	R/W	Data type	Remarks
0x8400	Internal alarm	1	NA	R	U16	see description in table 17
0x8401	Internal recovery fault 1	1	NA	R	U16	
0x8402	Internal recovery fault 2	1	NA	R	U16	
0x8403	Internal recovery fault 3	1	NA	R	U16	
0x8404	Internal recovery fault 4	1	NA	R	U16	
0x8405	Internal recovery fault 5	1	NA	R	U16	
0x8406	Internal recovery fault 6	1	NA	R	U16	
0x8407	Internal recovery fault 7	1	NA	R	U16	
0x8408	Internal permanent fault	1	NA	R	U16	
0x8409	ARMC fault	1	NA	R	U16	

4.6 Holding register table -AC Protection Parameter Area

Table 9 AC Protection Parameter Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0000	AC voltage overvoltage level 1 protection voltage threshold percentage	0.01	%	S16	R/W	10000	15000		
0x0001	AC voltage overvoltage level 1 protection time	0.01	s	S16	R/W	0	32767		
0x0002	AC voltage overvoltage level 2 protection voltage threshold percentage	0.01	%	S16	R/W	10000	15000		
0x0003	AC voltage overvoltage level 2 protection time	0.01	s	S16	R/W	0	32767		
0x0004	AC voltage overvoltage level 3 protection voltage threshold percentage	0.01	%	S16	R/W	10000	15000		

0x0005	AC voltage overvoltage level 3 protection time	0.01	s	S16	R/W	0	32767		
0x0006	AC voltage undervoltage level 1 protection voltage threshold percentage	0.01	%	S16	R/W	3000	10000		
0x0007	AC voltage undervoltage level 1 protection time	0.01	s	S16	R/W	0	32767		
0x0008	AC voltage undervoltage level 2 protection voltage threshold percentage	0.01	%	S16	R/W	3000	10000		
0x0009	AC voltage undervoltage level 2 protection time	0.01	s	S16	R/W	0	32767		
0x000A	AC voltage undervoltage level 3 protection voltage threshold percentage	0.01	%	S16	R/W	3000	10000		
0x000B	AC voltage undervoltage level 3 protection time	0.01	s	S16	R/W	0	32767		
0x000C	AC voltage recovery maximum percentage	0.01	%	S16	R/W	8000	15000		
0x000D	AC voltage recovery minimum percentage	0.01	%	S16	R/W	2000	10000		
0x000E	AC voltage recovery time	0.01	s	S16	R/W	0	32767		
0x000F	AC voltage overfrequency level 1 protection	0.01	Hz	S16	R/W	5000	6600		
0x0010	AC voltage overfrequency level 1 protection time	0.01	s	S16	R/W	0	32767		
0x0011	AC voltage overfrequency level 2 protection	0.01	Hz	S16	R/W	5000	6600		
0x0012	AC voltage overfrequency level 2 protection time	0.01	s	S16	R/W	0	32767		
0x0013	AC voltage overfrequency level 3 protection	0.01	Hz	S16	R/W	5000	6600		
0x0014	AC voltage overfrequency level 3 protection time	0.01	s	S16	R/W	0	32767		
0x0015	AC voltage underfrequency level 1 protection	0.01	Hz	S16	R/W	4500	6000		
0x0016	AC voltage underfrequency level 1 protection time	0.01	s	S16	R/W	0	32767		
0x0017	AC voltage underfrequency level 2 protection	0.01	Hz	S16	R/W	4500	6000		
0x0018	AC voltage underfrequency level 2 protection time	0.01	s	S16	R/W	0	32767		

0x0019	AC voltage underfrequency level 3 protection	0.01	Hz	S16	R/W	4500	6000		
0x001A	AC voltage underfrequency level 3 protection time	0.01	s	S16	R/W	0	32767		
0x001B	AC frequency recovery maximum value	0.01	Hz	S16	R/W	4500	6600		
0x001C	AC frequency recovery minimum value	0.01	Hz	S16	R/W	4500	6000		
0x001D	AC frequency recovery time	0.01	s	S16	R/W	0	32767		
0x001E	Moving average upper voltage protection value	0.1	V	S16	R/W	0	32767		
0x001F	Moving average upper voltage protection time	0.01	s	S16	R/W	0	32767		
0x0020	Moving average lower voltage protection value	0.1	V	S16	R/W	0	32767		
0x0021	Moving average lower voltage protection time	0.01	s	S16	R/W	0	32767		
0x0022	Moving average recovery time	0.01	s	S16	R/W	0	32767		
0x0023	Mains voltage imbalance	0.01	%	S16	R/W	1	1000		
0x0024	AC voltage overvoltage level 1 protection	0.1	V	S16	R/W	0	32767		
0x0025	AC voltage overvoltage level 2 protection	0.1	V	S16	R/W	0	32767		
0x0026	AC voltage overvoltage level 3 protection	0.1	V	S16	R/W	0	32767		
0x0027	AC voltage undervoltage level 1 protection	0.1	V	S16	R/W	0	32767		
0x0028	AC voltage undervoltage level 2 protection	0.1	V	S16	R/W	0	32767		
0x0029	AC voltage undervoltage level 3 protection	0.1	V	S16	R/W	0	32767		
0x002A	AC voltage recovery maximum value	0.1	V	S16	R/W	0	32767		
0x002B	AC voltage recovery minimum value	0.1	V	S16	R/W	0	32767		

4.7 Holding register table -Active Power Derating Parameter Area

Table 10 Active Power Derating Parameter Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0100	Grid overvoltage derating trigger value	0.1	V	S16	R/W	0	32767		
0x0101	Grid overvoltage derating recovery value	0.1	V	S16	R/W	0	32767		
0x0102	Grid overvoltage derating slope	0.10	%	S16	R/W	1	1000		
0x0103	Grid overvoltage derating filtering time	1	s	S16	R/W	1	90		
0x0104	Curve 1 Overfrequency load drop protection starting point	0.01	Hz	S16	R/W	5000	7200		
0x0105	Curve 1 Overfrequency load drop protection end point	0.01	Hz	S16	R/W	5000	7200		
0x0106	Curve 1 Overfrequency load reduction rate	0.01	%	S16	R/W	1	10000		
0x0107	Curve 1 Overfrequency download to restore frequency value	0.01	Hz	S16	R/W	4900	6600		
0x0108	Curve 1 Overfrequency download recovery time	1	s	S16	R/W	0	1200		
0x0109	Curve 1 Underfrequency load drop starting point	0.01	Hz	S16	R/W	4500	6000		
0x010A	Curve 1 Underfrequency load drop end point	0.01	Hz	S16	R/W	5000	7200		
0x010B	Curve 1 Underfrequency load reduction rate	0.01	%	S16	R/W	1	10000		
0x010C	Curve 1 Underfrequency download to restore frequency value	0.01	Hz	S16	R/W	4500	6600		
0x010D	Curve 1 Underfrequency download recovery time	1	s	S16	R/W	0	1200		
0x010E - 0x0110	-								
0x0111	Overfrequency Curve Selection	1		S16	R/W	0	1		
0x0112	Underfrequency curve selection	1		S16	R/W	0	1		
0x0113	Curve 2 Overfrequency load drop protection starting point	0.01	Hz	S16	R/W	4500	6000		
0x0114	Curve 2 Overfrequency load drop protection end point	0.01	Hz	S16	R/W	5000	7200		
0x0115	Curve 2 Overfrequency load reduction rate	0.01	%	S16	R/W	1	10000		

0x0116	Curve 2 Overfrequency download to restore frequency value	0.01	Hz	S16	R/W	4500	6600		
0x0117	Curve 2 Overfrequency download recovery time	1	s	S16	R/W	0	1200		
0x0118	Curve 2 Underfrequency load drop starting point	0.01	Hz	S16	R/W	4500	6000		
0x0119	Curve 2 Underfrequency load drop end point	0.01	Hz	S16	R/W	5000	7200		
0x011A	Curve 2 Underfrequency load reduction rate	0.01	%	S16	R/W	1	10000		
0x011B	Curve 2 Underfrequency download to restore frequency value	0.01	Hz	S16	R/W	4500	6600		
0x011C	Curve 2 Underfrequency download recovery time	1	s	S16	R/W	0	1200		

4.8 Holding register table -Reactive Power Derating Parameter Area

Table 11 Reactive Power Derating Parameter Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0200	PF(P) setting	0.1	%	S16	R/W	-1000	1000		Need to set Reactive power mode setting(0x0601) to 3: Local PF setting before use
0x0201	PF(P) curve P1	0.1	%	S16	R/W	0	1000		
0x0202	PF(P) curve PF1	0.1	%	S16	R/W	-1000	1000		
0x0203	PF(P) curve P2	0.1	%	S16	R/W	0	1000		
0x0204	PF(P) curve PF2	0.1	%	S16	R/W	-1000	1000		
0x0205	PF(P) curve trigger voltage	0.1	V	S16	R/W	0	32767		
0x0206	PF(P) curve reset voltage	0.1	V	S16	R/W	0	32767		
0x0207	Q(U) curve U1	0.1	V	S16	R/W	0	32767		
0x0208	Q(U) curve Q1	0.1	%	S16	R/W	-1000	1000		
0x0209	Q(U) curve U2	0.1	V	S16	R/W	0	32767		
0x020A	Q(U) curve Q2	0.1	%	S16	R/W	-1000	1000		

0x020B	Q(U) curve U1i	0.1	V	S16	R/W	0	32767		
0x020C	Q(U) curve Q1i	0.1	%	S16	R/W	-1000	1000		
0x020D	Q(U) curve U2i	0.1	V	S16	R/W	0	32767		
0x020E	Q(U) curve Q2i	0.1	%	S16	R/W	-1000	1000		
0x020F	Q(U) curve trigger power	0.1	%	S16	R/W	50	1000		
0x0210	Q(U) curve reset power	0.1	%	S16	R/W	50	1000		
0x0211	Q(P) curve 1	0.1	%	S16	R/W	50	1000		
0x0212	Q(P) curve 2	0.1	%	S16	R/W	50	1000		
0x0213	Q(P) curve 3	0.1	%	S16	R/W	50	1000		
0x0214	Q(P) curve 4	0.1	%	S16	R/W	50	1000		
0x0215	Q(P) curve 5	0.1	%	S16	R/W	50	1000		
0x0216	Q(P) curve 6	0.1	%	S16	R/W	50	1000		

4.9 Holding register table -Other Parameter Area

Table 12 Other Parameter Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0500	Reconnection time	1	s	S16	R/W	1	1200		
0x0501	PV start-up voltage	1	V	S16	R/W	180	400		
0x0502	PV Power Mutation Slow Start Slope (HECO)	0.01	%	S16	R/W	1	10000		
0x0503	Soft start power step after utility power error	0.01	%	S16	R/W	1	10000		
0x0504	Normal soft-stop power step	0.01	%	S16	R/W	1	10000		
0x0505	Normal soft-start power step	0.01	%	S16	R/W	1	10000		
0x0506	Normal load reduction step size	0.01	%	S16	R/W	1	10000		
0x0507	Minimum start-up temperature	0.1	℃	S16	R/W	-400	1000		

0x0508	The module over-temperature protection error point	0.1	℃	S16	R/W	0	1500		
0x0509	Environmental over-temperature protection error point	0.1	℃	S16	R/W	0	1500		
0x050A	HVRT trigger voltage	0.1	%	S16	R/W	1000	1500		
0x050B	LVRT trigger voltage	0.1	%	S16	R/W	700	1000		
0x050C	LVRT positive sequence reactive current coefficient	0.1	%	S16	R/W	0	3000		
0x050D	LVRT negative sequence reactive current coefficient	0.1	%	S16	R/W	0	3000		
0x050E	Active power deloading percentage setting value	0.1	%	S16	R/W	0	1000		Need to set Active power mode setting(0x0602) to 2: Local derating setting before use
0x050F	Reactive power percentage setting value	0.1	%	S16	R/W	-1000	1000		Need to set Reactive power mode setting(0x0601) to 2: Local Q setting before use
0x0510 - 0x0519									
0x051A	Apparent power derating percentage setting value	0.1	%	S16	R/W	0	1000		
0x051B	Export limit mode setting	1		S16	R/W	0	2	0-Off 1-On	
0x051C	Export limit power setting	0.1	%	S16	R/W	-1000	1000		
0x051D	Export limit control mode	1		S16	R/W	1	2	1-Per total (phase average) 2-Per phase (lowest phase)	
0x051E	Low penetration reactive power limit setting	0.1	%	S16	R/W	-1000	1000		
0x051F	-								
0x0520	Export limit failure protection enable	1		S16	R/W	0	1		
0x0521	Export limit failure power setting	0.1	%	S16	R/W	0	1000		

0x0524	Active power deloading fixed value	0.1	KW	S16	R/W	-32768	32767		Need to set Active power mode setting(0x0602) to 2: Local derating setting before use
0x0525	Reactive power deloading fixed value	0.1	KW	S16	R/W	-32768	32767		Need to set Reactive power mode setting(0x0601) to 2: Local Q(U) setting before use

4.10 Holding register table -Enable Control Parameter Area

Table 13 Enable Control Parameter Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0600	Controller parameter group selection	1		S16	R/W	0	100		For Debug
0x0601	Reactive power mode setting	1		S16	R/W	0	7	0: No adjustment 1: Remote Q/PF(P) setting (power off is not saved) 2: Local Q(U) setting 3: Local PF Settings 4. PF(P) curve setting 5. Q(U) curve setting 6. Set according to power 7. Q(P)	Remote: does not write EEPROM, suitable for power grid dispatching and other frequent regulation requirements; Local: written to EEPROM, do not frequently adjusted;
0x0602	Active power mode setting	1		S16	R/W	0	3	0: No adjustment 1: Remote derating setting (power is not saved) 2: Local derating setting 3: Set by power	
0x0603	Full MPPT scan enable setting	1		S16	R/W	0	1		
0x0604	-								

- 0x0607									
0x0608	LVRT mode setting	1		S16	R/W	0	4	0: Off 1:Zero current mode 2:Reactive power mode 3:Active power priority mode 4:Zero current threshold	
0x0609	HVRT mode setting	1		S16	R/W	0	2		
0x060A - 0x0624	-								
0x0625	Group string detection enabled	1		S16	R/W	0	1	0：off 1：on	
0x0626 - 0x062A									
0x062B	MRPF setting*(320K)*	1		S16	R/W	0	1	0：off 1：on	

4.11 Holding register table -Control Command Area

Table 14 Control Command Area

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0700	On-off command	1	Hex	S16	R/W	0	65535	on：0x5555 off：0x7777	
0x0701 - 0x0702	-								
0x0703	Automated testing (Italy)	1	Hex	S16	R/W	0	65535		
0x0704 - 0x0706	-								
0x0707	Remote PF setting	0.1	%	S16	R/W	-1000	1000		Need to set Reactive power mode setting(0x0601) to 1:Remote Q/PF(P) setting before use

0x0708	Remote active power setting	0.1	%	S16	R/W	0	1000		Need to set Active power mode setting(0x0602) to 1:Remote derating setting before use
0x0709	Remote reactive power setting	0.1	%	S16	R/W	-600	600		Need to set Reactive power mode setting(0x0601) to 1:Remote derating setting before use

4.12 Holding register table -Inverter Basic Information Area DSP

Table 15 Inverter Basic Information Area DSP

Address	NameEn	Ratio	Unit	Data Type	R/W	Min	Max	Remark	Instructions
0x0900	Grid Code	1		S16	R/W	0	36	3：CEI0-16; 4：EN50549-2; 5：CHILE; 6：EN50549-1; 7：CHINA-HV; 8：NB/T 32004 ; 9：IEC61727; 7：China_HV; 10：KS; 11：ABNT NBR 16149; 14：NRS-097-2-1; 16：VDE4105; 23：CEI-021; 31：ANRE;	
0x0901	N Line	1	Hex	S16	R/W	0	65535	There is an N-wire connection：5A5A，There is no N-wire connection：A5A5	
0x0902	-								
0x0903	Daily power generation	1	kWh	S16	R/W	0	65535		
0x0904	Daily power generation time	1	h	U16	R/W	0	6550		
0x0905	Total power generation (MSB)	1	kWh	U16	R/W	0	65535		

0x0906	Total power generation (LSB)	1	kWh	U16	R/W	0	65535		
0x0907	Total power generation time(MSB)	1	h	U16	R/W	0	65535		
0x0908	Total power generation time (LSB)	1	h	U16	R/W	0	65535		
0x0909 - 0x090B	-								
0x090C	Rated power	1	kW	U16	R	0	65535		
0x090E	Month Energy (MSB)	1	kWh	S16	R/W	0	65535		
0x090F	Month Energy (LSB)	1	kWh	U16	R/W	0	65535		
0x0910	Year Energy (MSB)	1	kWh	U16	R/W	0	65535		
0x0911	Year Energy (LSB)	1	kWh	U16	R/W	0	65535		
0x0912	Week Energy(MSB)	1	kWh	U16	R/W	0	65535		
0x0913	Week Energy (LSB)	1	kWh	U16	R/W	0	65535		

4.13 Holding register table -Basic Information area for ARM

Table 16 Nverter Basic Information Area ARM

Address	NameEn	Ratio	Unit	DataType	R/W	Min	Max	Remark	Instructions
0x0B02	DSPM software version number	1		U16	R	0	65535		
0x0B03	ARMS software version number	1		U16	R	0	65535		
0x0B04	ARDC software version number	1		U16	R	0	65535		
0x0B05	CPLD software version number	1		U16	R	0	65535		
0x0B07	Serial number 01~02 digits	1	STR	U16	R/W	0	65535		
0x0B08	Serial number 03~04 digits	1	STR	U16	R/W	0	65535		
0x0B09	Serial number 05~06 digits	1	STR	U16	R/W	0	65535		
0x0B0A	Serial number 07~08 digits	1	STR	U16	R/W	0	65535		
0x0B0B	Serial number 09~10 digits	1	STR	U16	R/W	0	65535		
0x0B0C	Serial number 11~12 digits	1	STR	U16	R/W	0	65535		
0x0B0D	Serial number 13~14 digits	1	STR	U16	R/W	0	65535		
0x0B0E	Serial number 15~16 digits	1	STR	U16	R/W	0	65535		
0x0B0F	Serial number 17~18 digits	1	STR	U16	R/W	0	65535		
0x0B10	Serial number 19~20 digits	1	STR	U16	R/W	0	65535		
0x0B11	The local mailing address	1		U16	R/W	1	247		

0x0B12	Device name 01-02 digits	1	STR	U16	R/W	0	65535		
0x0B13	Device name 03-04 digits	1	STR	U16	R/W	0	65535		
0x0B14	Device name 05-06 digits	1	STR	U16	R/W	0	65535		
0x0B15	Device name 07-08 digits	1	STR	U16	R/W	0	65535		
0x0B16	Device name 09-10 digits	1	STR	U16	R/W	0	65535		
0x0B17	Device name: 11-12 digits	1	STR	U16	R/W	0	65535		
0x0B18	Device name: 13-14 digits	1	STR	U16	R/W	0	65535		
0x0B19	Device name: 15-16 digits	1	STR	U16	R/W	0	65535		
0x0B1A	Device name: 17~18 digits	1	STR	U16	R/W	0	65535		
0x0B1B	Device name: 19~20 digits	1	STR	U16	R/W	0	65535		
0x0B24	RTC/year	1		U16	R/W	0	99		
0x0B25	RTC/month	1		U16	R/W	1	12		
0x0B26	RTC/day	1		U16	R/W	1	31		
0x0B27	RTC/hour	1		U16	R/W	0	23		
0x0B28	RTC/min	1		U16	R/W	0	59		
0x0B29	RTC/sec	1		U16	R/W	0	59		
0x0B42	PID function switch	1		U16	R/W	0	1		
0x0B43	PID output voltage	0.1	V	U16	R/W	5000	10000		
0x0B44	PID function startup time	1		U16	R/W	0	65535		for example:0x0852, In decimal terms it's 2130,That's 21:30
0x0B45	PID function off time	1		U16	R/W	0	65535		
0x0B52	Meter/tracker protocol setting	1		U16	R/W	0	65535	0: off 1: on	
0x0B54	Meter CT ratio	1		U16	R/W	1	65535		
0x0B60	Baud rate setting	1		U16	R/W	0	10	1: 9600 2: 19200 3: 38400 4: 115200	1

5. Fault Analysis

Table 17 fault analysis

Fault type	Fault ID	Bit	Name of fault
Internal recoverable fault	1	Bit0	External Fan Warning
	2	Bit1	Internal Fan Warning
	3	Bit2	
	4	Bit3	EEPROM R/W Warning
	5	Bit4	DC SPD Warning
	6	Bit5	Temp Sensor Warning
	7	Bit6	Boost CBC overcurrent*(320k)*
	8	Bit7	AC SPD Warning
	9	Bit8	
	10	Bit9	INV CBC overcurrent*(320k)*
	11	Bit10	
	12	Bit11	
	13	Bit12	
	14	Bit13	
	15	Bit14	
	16	Bit15	
Internal recoverable fault 1	17	Bit0	Bus Overvolt Err
	18	Bit1	
	19	Bit2	Bus Unbalance Err
	20	Bit3	Boost Sof start Overtime Err
	21	Bit4	Inv Softstart Overtime Err
	22	Bit5	
	23	Bit6	Boost 1 Curr High Err
	24	Bit7	Grid Line Vol Overvolt Err
	25	Bit8	Grid Phase Vol Overvolt Err
	26	Bit9	Inv Curr High Err
	27	Bit10	Grid Freq High Err
	28	Bit11	Grid Freq Low Err
	29	Bit12	No Utility Err
	30	Bit13	Relay Err
	31	Bit14	Over Temp Err
	32	Bit15	Inv Curr Offset Err
Internal recoverable	33	Bit0	Inv Vol Offset Err

fault 2	34	Bit1	Inv DCI Curr Offset Err
	35	Bit2	Inv DCI High Err
	36	Bit3	Iso Low Err
	37	Bit4	GFCI High Err
	38	Bit5	Freq Detect Err
	39	Bit6	
	40	Bit7	Slave MCU Communication Err
	41	Bit8	Hardware Inv Overcurr Err
	42	Bit9	Grid Volt unBalance Err
	43	Bit10	
	44	Bit11	Inv Curr unBalance Err
	45	Bit12	
	46	Bit13	
	47	Bit14	Hardware Bus Overvolt Err
	48	Bit15	GFCI CT Err
Internal recoverable fault 3	49	Bit0	
	50	Bit1	Internal Hardware Err
	51	Bit2	
	52	Bit3	PV2 Rev Connect Err
	53	Bit4	Boost 2 Curr High Err
	54	Bit5	PV2 Volt High Err
	55	Bit6	
	56	Bit7	Inv OpenLoop Self Check Err
	57	Bit8	
	58	Bit9	PV1 Rev Connect Err
	59	Bit10	PV1 Volt High Err
	60	Bit11	PV3 Rev Connect Err
	61	Bit12	PV3 Volt High Err
	62	Bit13	
	63	Bit14	
	64	Bit15	
Internal recoverable fault 4	65	Bit0	
	66	Bit1	Hardware Bst Overcurr Err
	67	Bit2	
	68	Bit3	
	69	Bit4	

	70	Bit5	
	71	Bit6	PV4 Volt High Err
	72	Bit7	PV4 Rev Connect Err
	73	Bit8	PV5 Volt High Err
	74	Bit9	PV5 Rev Connect Err
	75	Bit10	PV6 Volt High Err
	76	Bit11	PV6 Rev Connect Err
	77	Bit12	PV7 Volt High Err
	78	Bit13	PV7 Rev Connect Err
	79	Bit14	PV8 Volt High Err
	80	Bit15	PV8 Rev Connect Err
Internal recoverable fault 5	81	Bit0	Hardware 5V Low Err
	82	Bit1	GFCI Steady Err
	83	Bit2	
	84	Bit3	
	85	Bit4	Boost3 Curr High Err
	86	Bit5	Cap and Pll Detect Err
	87	Bit6	PV9 Volt High Err
	88	Bit7	PV9 Rev Connect Err
	89	Bit8	
	90	Bit9	
	91	Bit10	
	92	Bit11	
	93	Bit12	
	94	Bit13	
	95	Bit14	
	96	Bit15	
Internal recoverable fault 6	97	Bit0	Boost 4 Curr High Err
	98	Bit1	Boost 5 Curr High Err
	99	Bit2	Boost 6 Curr High Err
	100	Bit3	Boost 7 Curr High Err
	101	Bit4	Boost 8 Curr High Err
	102	Bit5	Boost 9 Curr High Err
	103	Bit6	DSPM ARMS Communication Fault
	104	Bit7	
	105	Bit8	
	106	Bit9	
	107	Bit10	
	108	Bit11	
	109	Bit12	
	110	Bit13	
	111	Bit14	

	112	Bit15	
Internal recoverable fault 7	113	Bit0	
	114	Bit1	
	115	Bit2	
	116	Bit3	
	117	Bit4	
	118	Bit5	
	119	Bit6	
	120	Bit7	
	121	Bit8	
	122	Bit9	
	123	Bit10	
	124	Bit11	
	125	Bit12	
	126	Bit13	
	127	Bit14	
	128	Bit15	
Permanent internal failure	129	Bit0	Bus Overvolt Fault
	130	Bit1	
	131	Bit2	Bus unBalance Fault
	132	Bit3	Relay Fault
	133	Bit4	GFCI Steady Fault
	134	Bit5	Hardware Boost Overcurr Fault
	135	Bit6	DCI High Fault
	136	Bit7	Hardware Bus Overvolt Fault
	137	Bit8	Hardware Inv Overcurr Fault
	138	Bit9	
	139	Bit10	
	140	Bit11	
	141	Bit12	
	142	Bit13	Internal Hardware Fault
	143	Bit14	Inv OpenLoop Self Check Fault
	144	Bit15	
ARMC fault	145	Bit0	
	146	Bit1	PID Hardware Fault
	147	Bit2	PID BUS Voltage Balance Fault
	148	Bit3	PID Load Break Fault
	149	Bit4	PID Load Short Fault
	150	Bit5	ARMC EEPROM Read Fault

	151	Bit6	ARMC EEPROM Write Fault
	152	Bit7	ARMC EEPROM Checksum Fault
	153	Bit8	RTC Hardware Fault
	154	Bit9	SPD Hardware Fault
	155	Bit10	NOR-FLASH Hardware Fault
	156	Bit11	ARMS Upgrade Failed
	157	Bit12	DSPM Upgrade Failed
	158	Bit13	CPLD Upgrade Failed
	159	Bit14	
	160	Bit15	

6. CRC16-RTU verification code

```
const uint8 chCRCHTalbe[] =
{
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40
};
const uint8 chCRCLTalbe[] =
{
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06, 0x07, 0xC7,
0x05, 0xC5, 0xC4, 0x04, 0xCC, 0x0C, 0x0D, 0xCD, 0x0F, 0xCF, 0xCE, 0x0E,
0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09, 0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9,
0x1B, 0xDB, 0xDA, 0x1A, 0x1E, 0xDE, 0xDF, 0x1F, 0xDD, 0x1D, 0x1C, 0xDC,
```

```
0x14, 0xD4, 0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32,
0x36, 0xF6, 0xF7, 0x37, 0xF5, 0x35, 0x34, 0xF4, 0x3C, 0xFC, 0xFD, 0x3D,
0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A, 0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38,
0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF,
0x2D, 0xED, 0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60, 0x61, 0xA1,
0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67, 0xA5, 0x65, 0x64, 0xA4,
0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F, 0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB,
0x69, 0xA9, 0xA8, 0x68, 0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA,
0xBE, 0x7E, 0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71, 0x70, 0xB0,
0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97,
0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C, 0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E,
0x5A, 0x9A, 0x9B, 0x5B, 0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89,
0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83,
0x41, 0x81, 0x80, 0x40
};

uint16 Modbus_CRC16(const uint8 * pchMsg, uint16 uDataLen)
{
    uint8 uchCRCHi = 0xFF;
    uint8 uchCRCLo = 0xFF;
    uint16 uIndex;
    while (uDataLen--)
    {

        uIndex = uchCRCLo ^ *pchMsg++ ;
        uchCRCLo = uchCRCHi ^ chCRCHTalbe[uIndex];
        uchCRCHi = chCRCLTalbe[uIndex];
    }
    // ModBus CRC16 Format is "uchCRCLo + uchCRCHi".
    return (((uint16)( uchCRCLo) << 8) | uchCRCHi);
}
```

7. Modbus address assignment

When the communication address is 0, it is a broadcast instruction. Each device in the subnet needs to report the current device sub-address (0x64 by default) and SN. The master assigns a communication address to each device according to the received device SN to ensure that the communication address of the device in the subnet is unique and not duplicate.

7.1 Address Search Command(Broadcast)

The master sends the broadcast network instruction and receives the SN and communication address of each device in the subnet. The device response time adopts random number response, and the maximum response time is 3s.

Sending format	Modbus address (1Byte)	Function code (1Byte)	Start address (2Bytes)	Register number (2Bytes)		CRC16 (2Bytes)
Example	0x00	0x35	0x0B 0x07	0x00	0x0B	CRCL CRCH
Response format	**Modbus address (1Byte)**	**function code (1Byte)**	**number of bytes (1Byte)**	**Subdevice SN (20Bytes)**	**Sub-device communication address (2Bytes)**	**CRC16 (2Bytes)**
Example	0x64	0x35	0x16	XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	0x00 0x64	CRCL CRCH

7.2 Address Assignment Command

The master assigns communication addresses to each device in the subnet through broadcast instructions, and the sub-devices identify and set their own sub-addresses through SN. The devices whose addresses have been set do not reply to the broadcast networking instructions in 7.1. The procedure is described as follows:

Sending format	Modbus address (1Byte)	Function code (1Byte)	Start address (2Bytes)	Register number (2Bytes)	CRC16 (2Bytes)	Sending format
example	0x00	0x38	0x0B 0x07	XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	0x00 0x02	CRCL CRCH
Response format	**Modbus address (1Byte)**	**function code (1Byte)**	**number of bytes (1Byte)**	**Subdevice SN (20Bytes)**	**Sub-device communication address (2Bytes)**	**CRC16 (2Bytes)**
example	0x02	0x38	0x16	XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX	0x00 0x02	CRCL CRCH

				XX XX XX XX		

7.3 Unregistering Address Command(Broadcast)

By using this command, the host computer can remove all devices from the communication network and restore their addresses to the default value (0x64). The device whose address has been deregisted can still reply to the broadcast network instruction 7.1. The device does not need to reply to this instruction.

Sending format	**Modbus address (1Byte)**	**Function code (1Byte)**	**Start address (2Bytes)**	**Fixed value (2Bytes)**	**CRC16 (2Bytes)**
example	0x00	0x39	0x0B 0x07	0x00 0x64	CRCL CRCH
Response format	**Modbus address (1Byte)**	**function code (1Byte)**	**number of bytes (1Byte)**	**fixed value (2Bytes)**	**CRC16 (2Bytes)**
No response required					

8. Example

8.1 Read one input register instruction.

Assuming the slave address is 1, need to obtain data for address 0x8000.
Host Send (HEX):
01 04 80 00 00 01 18 0A
Slave Response (HEX):
01 04 02 0E D8 BD 0A
Data analysis: The value of Line voltage Uab is 380.0V

8.2 Read multiple input registers instruction

Assuming the slave address is 1, need to obtain data for address 0x8000,0x8001,0x8002.
Host Send (HEX):
01 04 80 00 00 03 99 CB
Slave Response (HEX):

01 04 06 15 50 15 46 15 38 48 C1
Data analysis: The value of Line voltage Uab is 545.6V,Ubc is 544.6V,Uca is 543.2V.

8.3 Read one hold register instruction

Assuming the slave address is 1, need to obtain data for address 0x0B24.
Host Send (HEX):
01 03 0B 24 00 01 C6 25
Slave Response (HEX):
01 03 02 00 15 79 8B
Data analysis: The RTC/year is 21.

8.4 Read multiple hold registers instruction

Assuming the slave address is 1, need to obtain data for address 0x0B24,0x0B25,0x0B26,0x0B27.
Host Send (HEX):
01 03 0B 24 00 04 06 26
Slave Response (HEX):
01 03 08 00 15 00 02 00 04 00 07 A8 D5
Data analysis: The year is 21,the month is 2,the day is 4,the hour is 7.

8.5 Write a hold register instruction

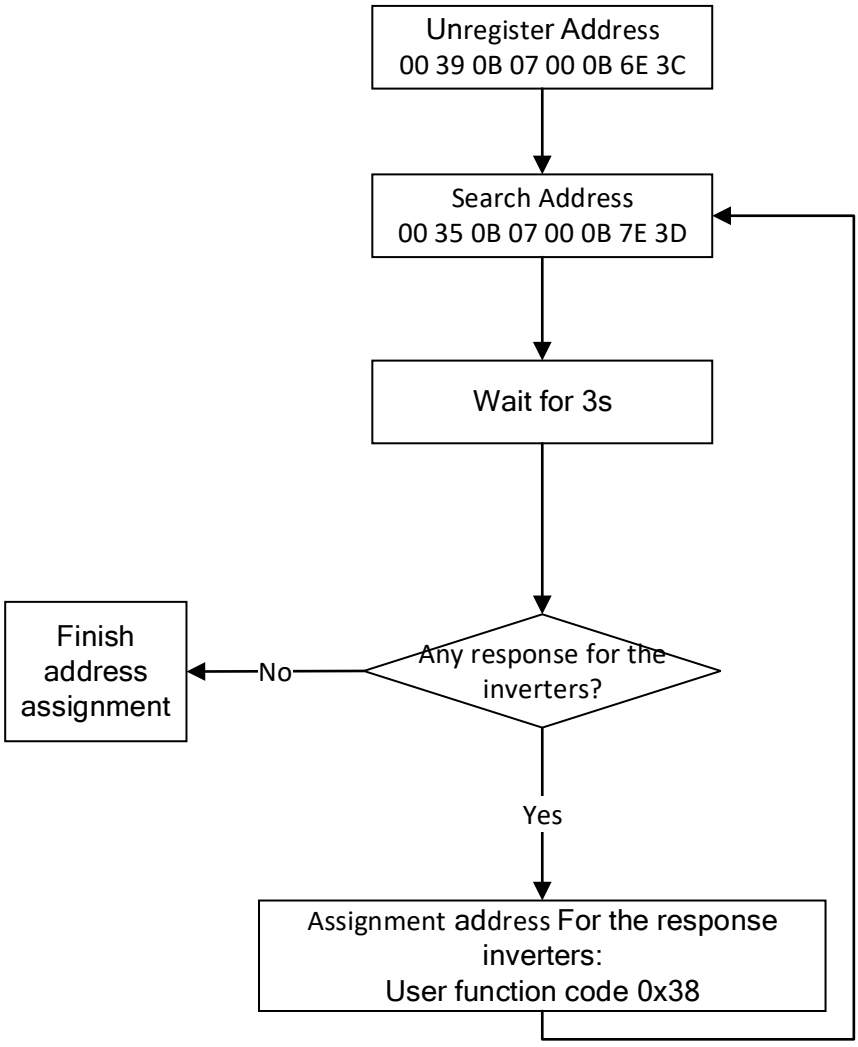
Assuming the slave address is 1, need to obtain data for address 0x0B24.
Host Send (HEX):
01 06 0B 24 00 17 8B EB
Slave Response (HEX):
01 06 0B 24 00 17 8B EB
Data analysis: The RTC/year was changed to 23.

8.6 Write multiple hold registers instruction

Assuming the slave address is 1, need to obtain data for address 0x0B24,0x0B25,0x0B26,0x0B27.
Host Send (HEX):
01 10 0B 24 00 04 08 00 17 00 0A 00 0C 00 10 E2 4E
Slave Response (HEX):
01 10 0B 24 00 04 83 E5
Data analysis: The year is 23,the month is 10,the day is 12,the hour is 16.

8.7 Modbus address assignment

8.7.1 Flow chart



8.7.2 Example

Step1:Unregister Address (HEX), Suggest sending 3 times

00 39 0B 07 00 0B 6E 3C
00 39 0B 07 00 0B 6E 3C
00 39 0B 07 00 0B 6E 3C

Step2:Send Search Address (HEX):

00 35 0B 07 00 0B 7E 3D

Step3: Wait for 3s…

Step4:The inverters response the SN :

Inverter 1: 64 35 16 53 4E 30 31 32 34 34 35 39 39 38 39 30 30 34 37 30 30 30 30 00 64 26 FB

Inverter 2: 64 35 16 53 4E 30 31 32 34 34 35 39 39 38 39 38 30 34 37 36 30 30 30 00 64 A7 77

Step5:Address Assignment：

Inverter 1: 00 38 0B 07 53 4E 30 31 32 34 34 35 39 39 38 39 30 30 34 37 30 30 30 30 00 02 37 C0

The inverter returns：

Inverter 1: 02 38 16 53 4E 30 31 32 34 34 35 39 39 38 39 30 30 34 37 30 30 30 30 00 02 6D 53

Address Assignment：

Inverter 2: 00 38 0B 07 53 4E 30 31 32 34 34 35 39 39 38 39 38 30 34 37 36 30 30 30 00 03 77 8C

The inverter returns：

Inverter 2: 03 38 16 53 4E 30 31 32 34 34 35 39 39 38 39 38 30 34 37 36 30 30 30 00 03 79 E6

Step6:Search Address again:

00 35 0B 07 00 0B 7E 3D

Wait for 3s…

If there is no inverter response, then finish. If there are other inverters response, go to the step5.